

Brian Smiley

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Employment & Experience

Arbor, San Francisco/NYC/remote

2025-Present

Head of Operations & Engineering - First Employee

- Ran end-to-end delivery of Arbor's Quantitative Trading Bootcamp including: venue, catering, budget, staff scheduling and accommodations, lesson planning, live digital platform troubleshooting, and instruction
- Built and maintain Arbor's simulated trading exchange platform that runs the Trading Bootcamp's live trading exercises
- Ran web development and event operations for Metagame, a 200-300 attendee game design conference (2025, 2026)
- Drafted, negotiated, and closed corporate training and vendor contracts
- Built the company's finance and HR infrastructure from zero: Xero bookkeeping, Stripe/Mercury/Gusto integrations for revenue and payroll tracking, cryptocurrency accounting, and business tax preparation

TypeSafe AI, San Francisco, CA

March 2025-December 2025

Full Stack Software Engineer - early engineer

- Built and maintained internal web platform for AI model research and evals
- Wrote pipelines for programmatic generation of datasets used in frontier model training

Arch Laboratories, New York, NY

April 2023-March 2025

Data Operations Analyst - Part Time

- Processed financial and document updates to clients' alternative investments on Arch's proprietary system

Schiminovich Astronomy & Instrumentation Laboratory, New York, NY

December 2016-June 2022

Lab Assistant: PI, Professor David Schiminovich, PhD

- Designed parts and assemblies of structural and optical components for Circumgalactic Hydrogen-Alpha Spectrograph (CHaS) in Solidworks, including original hardware designs and finite element analysis simulations
- Manufactured parts in Columbia University machine shop/Maker Space
- Interfaced with vendors and machine shops to purchase components and manufacture custom parts
- Assembled CHaS instrument components and assisted in engineering runs at Kitt Peak National Observatory

Columbia University Astrophysics, New York, NY

Summer 2012-Spring 2016

Lab Assistant: PI, Assistant Professor Bradley Johnson, PhD

- Designed parts and assemblies for cosmic microwave background (CMB) polarimetry experiments using Solidworks
- Operated cryostat hardware, including vacuum pumps and cryocoolers for testing experimental detectors
- Created schematic drawings for part manufacture, and interpreted others' drawings for reverse engineered digital mockups
- Designed, assembled, and operated experimental cryostat chamber for testing a superconducting magnetic bearing

Proficiencies

Stack: TypeScript, React, Next.js, Node.js, PostgreSQL, Prisma, Supabase, Vercel

AI Tooling: Claude Code, Codex, agentic workflows, MCP integration

Deployment: Netlify, Render, Vercel

Hardware/CAD: Solidworks, machining, 3d printing

Education

Columbia University BA, Astrophysics

May 2016

Coursera, Nand2Tetris, Parts 1 & 2

October 2021

Fractal Dev Accelerator, Summer 2024 cohort, 700 dev hours

June-August 2024

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Publications

Melso, N., Schiminovich, D., Smiley, B., Ong, H., Cevallos Aleman, I., and Stelea, I., “*The Circumgalactic H-alpha Spectrograph: First Light Observations and Data Analysis of Early Science Targets Featuring NGC 4631*”, vol. 53, no. 1, 2021.

Melso, N., Schiminovich, D., Smiley, B., Ong, H., Santiago, B., Sitaram, "Design and commissioning of the Circumgalactic H-alpha Spectrograph," Proc. SPIE 11447, Ground-based and Airborne Instrumentation for Astronomy VIII, 114470B (13 December 2020); <https://doi.org/10.1117/12.2561674>

Hamden, Erika T., et al. "FIREBall-2: advancing TRL while doing proof-of-concept astrophysics on a suborbital platform." Micro-and Nanotechnology Sensors, Systems, and Applications XI. Vol. 10982. International Society for Optics and Photonics, 2019.

Johnson, B. R., Columbro, F., Araujo, D., Limon, M., Smiley, B., Jones, G., ... & Gupta, S. (2017). "A large-diameter hollow-shaft cryogenic motor based on a superconducting magnetic bearing for millimeter-wave polarimetry." Review of Scientific Instruments, 88(10), 105102.

B. R. Johnson, P. A. R. Ade, D. Araujo, K. J. Bradford, D. Chapman, P. K. Day, J. Didier, S. Doyle, H. K. Eriksen, D. Flanigan, C. Groppi, S. Hillbrand, G. Jones, M. Limon, P. Mauskopf, H. McCarrick, A. Miller, T. Mroczkowski, B. Reichborn-Kjennerud, B. Smiley, J. Sobrin, I. K. Wehus: "The Detector System for the Stratospheric Kinetic Inductance Polarimeter (SKIP)", 2013, Journal of Low Temperature Physics: Volume 176, Issue 5 (2014), Page 741-748; arXiv:1308.0235. DOI: 10.1007/s10909-013-1014-3.

D. C. Araujo, P. A. R. Ade, J. R. Bond, K. J. Bradford, D. Chapman, G. Che, P. K. Day, J. Didier, S. Doyle, H. K. Eriksen, D. Flanigan, C. E. Groppi, S. N. Hillbrand, B. R. Johnson, G. Jones, M. Limon, A. D. Miller, P. Mauskopf, H. McCarrick, T. Mroczkowski, B. Reichborn-Kjennerud, B. Smiley, J. Sobrin, I. K. Wehus: "A LEKID-based CMB instrument design for large-scale observations in Greenland", 2014, Proc. SPIE. 9153, Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy VII, 91530W. (August 19, 2014); arXiv:1407.6249. DOI: 10.1117/12.2056828.

Sean Bryan, Kristi Bradford, George Che, Peter Day, Daniel Flanigan, Bradley R. Johnson, Glenn Jones, Bjorn Kjellstrand, Michele Limon, Philip Mauskopf, Heather McCarrick, Amber Miller, Brian Smiley: "Design of Dual-Polarization Horn-Coupled Kinetic Inductance Detectors for Cosmic Microwave Background Polarimetry", In 26th International Symposium on Space Terahertz Technology, ISSTT 2015; arXiv:1503.04684.